

Computer skills (Theory)

first part:

Hardware and Software

Note: things in this **color** are not necessary , but is good if you want to understand the topic better.

1.1 IT, software and hardware terms

what is a computer? (not in the material)

a computer is a machine that can take inputs, processes it and then give it as an output.

Computers consists of two sections:

Hardware: they are the physical part of the computers. Like mouse , keyboard and speakers.

Software: A part of a computer that we can't touch. They are a bunch of commands that makes the computer works. Like the OS (operating system) .

IT (Information technology) : it is a general term that refers to the use of computers to manipulate data . Like processing it and storing it .

1.2 Types of computers

computers , in general, comes in different forms for different goals, a computer might be very cheap and portable , while others may be stronger in computing power, the list below is about the most famous types of computers :

-Mainframe:

capacity: very powerful computes often connected to many other PCs over a network .

Speed: much faster than a regular pc . Used for processing large amount of data .

Cost: very expensive , only affordable by companies .

typical users: large companies and banks .

-PC (personal computer)

capacity: large. Can handle multiple disks combined with a working memory (RAM) .

Speed: Fast , normally measured in GHz.

Cost: cheap, and getting cheaper by the day .

Typical users : Anyone who can use a computer .

-Networked PC

capacity: large. Can handle multiple disks combined with a working memory (RAM) .

Speed: Fast , normally measured in GHz.

Cost: needs a normal pc and an inexpensive networking card to be add to that pc.

Typical users: anyone who can work with a networked pc .

-Laptop

capacity : Large hard disk and RAM , but often less capacity than a PC for the same price.

Speed: slower than a pc in the same price, yet still fast .

Cost : more expensive than a pc for the same power, because the components should be compacted which is an expensive task to do .

Typical users: Business users and people on the move.

-PDA & Palmtop

capacity : much smaller compared to a PC .

Speed; Much less than a pc unless you payed a lot extra.

Cost: expensive compared to pc .

Typical users: mostly business users.

1.3 The main parts of a computer

The system unit: It is the box or the case the houses the computer's parts like the mother-board.

Within the system unit:

The mother board: it is a board that connects all the vital computer components together, like the CPU and RAM.

The CPU (central processing unit) : It is the brain of the computer which do the calculations and directing the computer's components . It's speed often measured in GHz .

RAM (Random access memory): it is a temporary storage unit that saves data temporarily before moving it to the hard disk , when you open your computer, The operating system will be loaded at the ram , and applications as well .

ROM-BIOS: it is a chip held in your computer's mother board , it contains software that is required to make your computer work with the operating system (OS) .

Hard disk: This Is where computer stores data and information permanently like the operating system and programs. It is mostly a disk that can hold up a certain storage .

Input devices:

keyboard: an input device used for entering information into the computer . Such as letters and numbers.

Mouse: used to control the cursor on the screen, clicking and dragging to interact with programs .

Scanners : used to turn the physical copy of a document or a paper into a digital copy .

Output devices :

monitor: it is a screen that displays what you are doing on the computers, showing you the output of the process that you were doing in the computer.

Printer: turn the digital copy of a document into a physical copy.

Additional parts:

Modems : these devices allowed you to connect to a network throw a telephone line.

Speakers: they turn the data from computer that was saved as a sound, into a sound outside the computer .

Ports:

Serial port & parallel port: it is a socket located at the back of your motherboard that enables you to connect certain items.

The serial port is commonly labeled as COM1 or COM2.

And the parallel port as LPT1 and LPT2 .

USB: A newer type of sockets that become the standard . It enables you to connect a wide range of devices like printers and scanners , etc. and it is faster and more versatile than serial and parallel ports.

Storage devices :

CD ROM/DVD drives : these drives read data from a DVDs and CDs , whom can store a lot more data than floppy disks .
Additional note: CDs regularly holds 700MG, DVDs holds 4.7GB.

Tape backup : Tape backup is a way to store lots of data safely on a tape . Often used by businesses to make copies of there whole compute system or important information .

ZIP disks: Zip disks are a better version of floppy disks, offering more space to store data and higher data transformation speeds .
It can hold up to 250MG.

1.1.4 Computer performance

Your computer or pc can be fast or slow depending on several factors. Some of them are related to the type of your computer , and others are related to how you are managing your computer. below are the most common factors to consider:

CPU speed: generally, the faster your cpu is , the faster you computer will work . Because it is related to how fast your computer can do the processing .

CPUs nowadays are measured in **GHz** , and most common good CPUs have the speed of **2GHz or more** .

RAM size : The bigger your RAM size is , the better your computer's performance will become .

It is more like having more space in your room , which will allow you to store multiple stuff without having any issues .

Hard Disk speed and storage: Your hard disk have a speed of reading data in the computer, the faster your hard disk at reading data, the faster your computer will be at accessing data . And the capacity also effect it's speed .

De-fragmenting files: Overtime , files on hard disk can become fragmented . Meaning they are split into smaller files across the hard disk . De-fragmenting those files will make your computer faster .

Free hard disk space: Having a good amount of your hard disk space as a free space is better for optimal performance because it will make going through files much easier for the computer .

Multitasking consideration : Run only the important files or programs at once, avoid running a lot of programs at the same time.

1.2 Hardware

The CPU: it is the main processing unit of a computer. It do the calculations and responsible for directing other parts of the computer. It has several main functions:

Calculations: The CPU performs all the arithmetic and logical calculations within the computer. This includes basic math operations as well as more complex calculations required for tasks like image processing or video editing

Logic Control: The CPU also controls the flow of data and instructions within the computer. It fetches instructions from memory, decodes them, and then executes them. This ensures that programs run correctly and efficiently.

Immediate Access Memory: The CPU has a small amount of built-in memory called cache. This memory is used to store frequently accessed data and instructions, allowing the CPU to retrieve them quickly and improve performance.

CPU Speed: The speed of the CPU is measured in megahertz (MHz) or gigahertz (GHz). A higher number indicates a faster CPU, which can execute instructions more quickly.

The speed of the CPU has increased dramatically over the years. Early PCs had CPU's that ran at a few MHz, while modern CPU's can run at several GHz.

Memory: Computer memory is where data and programs are stored for the CPU to access. There are different types of memory, each with its own function.

Types of Memory:

RAM (Random Access Memory): This is the main "working" memory of the computer. It's where the operating system, programs, and data are stored while the computer is running.

RAM is volatile, meaning its contents are lost when the computer is turned off.

ROM (Read-Only Memory): This type of memory is non-volatile, meaning its contents are retained even when the power is off. It's often used to store essential instructions for starting the computer, such as the BIOS (Basic Input/Output System).

Flash BIOS: This is a type of ROM that can be updated with new instructions. This allows for bug fixes or feature updates to the BIOS without replacing the entire chip.

Video (Graphics) Memory: This memory is specifically for storing the data that's displayed on the screen. Modern computers have dedicated video cards with their own memory to handle the demands of high-resolution graphics and video playback.

Memory Measurement:

Computer memory is measured in units like bits, bytes, kilobytes (KB), megabytes (MB), gigabytes (GB), and terabytes (TB)

Additional input and output devices (adding to the previous ones):

Input: Light pen, Joystick , Microphones , and Touch Pads.

Output: Projection devices , Speakers .

Input and output at the same time: some devices can be an input device and an output device at the same time. Such as a **printer** that has a **scanner** .

1.2.6 Storage devices

Internal hard disks:

speed: very fast, referred to as “Average access time” , it is measured in Milliseconds .

Capacity: Large, measured in GB (which is 1024 megabytes)

Cost: Cheap, and getting cheaper by the time.

External hard disks:

speed : Slower than a normal internal hard disk, paying more will grant you higher speeds .

Capacity: Large.

Costs: more expensive than internal hard disks.

What is the difference between Internal and External hard disks?

The **internal** disk is located inside your system unit.

The **External** is attached to the computer outside the system unit, grant you mobility .

CD Disks :

Speed: slower than a hard disk .

Capacity : 650 to 700 Megabytes .

Cost: CDs are getting cheaper by the time .

DVDs:

speed: faster than CD , slower than Hard disks.

Capacity : 4.7 to 17 Gigabytes .

Cost: cheap, but a bit higher than CDs.

Computer skills

Software Section

مهارات الحاسوب
موضوع: البرمجيات

البرمجيات 1.3 Software

أنواع البرمجيات 1.3.1 Types of Software

What is an Operating System?

The operating system is a special type of program which loads automatically when you start your computer. The operating system allows you to use the advanced features of a modern computer without having to learn all the details of how the hardware works.

ما هو نظام التشغيل؟

نظام التشغيل هو نوع خاص من البرامج التي يتم تحميلها تلقائيًا عند بدء تشغيل الكمبيوتر. يتيح لك نظام التشغيل استخدام الميزات المتقدمة لجهاز كمبيوتر حديث دون الحاجة إلى معرفة كل تفاصيل كيفية عمل الأجهزة أو القطع الصلبة.

What is an Application Program?

An application program is the type of program which you use once the operating system has been loaded into memory (RAM). Examples include word processing programs, spreadsheets, databases, games and graphics programs.

ما هو البرنامج التطبيقي؟

البرنامج التطبيقي هو نوع البرنامج الذي تستخدمه بمجرد تحميل نظام التشغيل في الذاكرة (RAM). تشمل الأمثلة برامج معالجة النصوص وجدول البيانات وقواعد البيانات والألعاب وبرامج الرسومات.

Why are new versions of software released year after year?

New releases have less bugs, greater flexibility and more features. Often if you wish to see the version of software you are using, you can click on the product's Help drop down menu, and then click on the About (or similar) command.

لماذا يتم إصدار إصدارات جديدة من البرامج عامًا بعد عام؟

تحتوي الإصدارات الجديدة على أخطاء أقل ومرونة أكبر وميزات أكثر. غالبًا إذا كنت ترغب في معرفة إصدار البرنامج الذي تستخدمه، فيمكنك النقر فوق القائمة المنسدلة "مساعدة" الخاصة بالمنتج، ثم النقر فوق الأمر "حول" (أو ما شابه).

1.3.2 Operating System Software برامج نظام التشغيل

What is an Operating System?

The operating system is a special type of program which loads automatically when you start your computer. The operating system allows you to use the advanced features of a modern computer without having to learn all the details of how the hardware works. There are a number of different types of operating system in common use. The most widely used on PCs today is Windows. Later, different releases of Windows were introduced. The good news is that later versions of Microsoft Windows look almost identical to each other. Other examples of operating systems which may be run on PCs are: OS/2, UNIX and Linux.

Other types of computers, such as those manufactured by Apple have a completely different operating system.

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في وقت لاحق، تم تقديم إصدارات مختلفة من Windows والخبر السار هو أن الإصدارات الأحدث من Microsoft Windows تبدو متطابقة تقريبًا مع بعضها البعض.

من الأمثلة الأخرى لأنظمة التشغيل التي يمكن تشغيلها على أجهزة الكمبيوتر الشخصية: OS/2 و UNIX و Linux .

تتمتع أنواع أخرى من أجهزة الكمبيوتر، مثل تلك التي تصنعها شركة Apple بنظام تشغيل مختلف تمامًا.

1.3.3 Applications Software برامج التطبيقات

What is an application program?

An application program is the type of program which you use once the operating system has been loaded. Examples include word-processing programs, spreadsheets, databases, games and graphics programs.

ما هو البرنامج التطبيقي؟

البرنامج التطبيقي هو نوع البرنامج الذي تستخدمه بمجرد تحميل نظام التشغيل. تشمل الأمثلة برامج معالجة النصوص وجداول البيانات وقواعد البيانات والألعاب وبرامج الرسومات.

Word processing

A word processing program allows you to produce letters, memos, etc., easily. You can easily mail merge a list of names and addresses to produce mass mailers, individually addressed to customers or subscribers.

معالجة النصوص

يتيح لك برنامج معالجة النصوص إنتاج الرسائل والمذكرات وما إلى ذلك بسهولة. يمكنك دمج قائمة بالأسماء والعناوين بسهولة لإنتاج رسائل بريدية مجمعة، موجهة بشكل فردي إلى العملاء أو المشتركين.

Spreadsheets

A spreadsheet program allows you to work out a company's income, expenditure and then calculate the balance. It enables you to make 'what if' type projections of how the company will fair in the future and to forecast how changes in prices will affect profits.

جداول البيانات

يتيح لك برنامج جداول البيانات حساب دخل الشركة ونفقاتها ثم حساب الرصيد. يمكنك من عمل توقعات من نوع "ماذا لو" حول أداء الشركة في المستقبل والتنبؤ بكيفية تأثير التغييرات في الأسعار على الأرباح.

Databases

A database program allows you to compile information and then to search this information to extract just the information you require. For instance, if you have a database of all the equipment housed within an office you can very simply produce a report listing only the equipment above a certain value.

قواعد البيانات

يتيح لك برنامج قاعدة البيانات تجميع المعلومات ثم البحث في هذه المعلومات لاستخراج المعلومات التي تحتاجها فقط. على سبيل المثال، إذا كان لديك قاعدة بيانات بجميع المعدات الموجودة داخل المكتب، فيمكنك ببساطة شديدة إنشاء تقرير يسرد فقط المعدات التي تزيد قيمتها عن قيمة معينة.

Presentation

A presentation program allows you to produce professional-looking presentations, which can be printed out directly onto slides for use with an overhead projector. Alternatively, you can display your presentations directly on a computer screen or via a computerised projector.

العرض التقديمي

يتيح لك برنامج العرض التقديمي إنتاج عروض تقديمية ذات مظهر احترافي، والتي يمكن طباعتها مباشرة على شرائح لاستخدامها مع جهاز عرض علوي. بدلاً من ذلك، يمكنك عرض عروضك التقديمية مباشرة على شاشة الكمبيوتر أو عبر جهاز عرض محوسب.

Accounts/Payroll

In most large organisations, the accounts are maintained by a computerised system. Due to the repetitive nature of accounts, a computer system is ideally suited to this task and accuracy is guaranteed.

الحسابات / الرواتب

في معظم المؤسسات الكبيرة، يتم الاحتفاظ بالحسابات بواسطة نظام محوسب. نظرًا للطبيعة المتكررة للحسابات، فإن نظام الكمبيوتر مناسب بشكل مثالي لهذه المهمة والدقة مضمونة.

Web browsing

Applications used to view and interact with the World Wide Web (WWW).

تصفح الويب (الأنترنت)

هي التطبيقات المستخدمة لعرض والتفاعل مع شبكة الويب العالمية (WWW).

Web authoring

These applications allow almost anyone to create a web site, quickly and easily.

تأليف الويب

تتيح هذه التطبيقات لأي شخص تقريبًا إنشاء موقع ويب بسرعة وسهولة.

واجهة المستخدم الرسومية (GUI) Graphical User Interface (GUI) 1.3.4

1.3.4.1 Understanding the term Graphical User Interface (GUI).

فهم مصطلح واجهة المستخدم الرسومية

What is a Graphical User Interface?

A Graphical User Interface (GUI) is simply an additional part of the operating system which displays windows and drop-down menus, and also enables you to drive your computer using a mouse. Examples of operating systems which use a GUI include Windows and IBM's OS/2.

If you used an old operating system like the original version of DOS, there was no GUI.

ما هي واجهة المستخدم الرسومية؟

واجهة المستخدم الرسومية (GUI) هي ببساطة جزء إضافي من نظام التشغيل الذي يعرض النوافذ والقوائم المنسدلة، كما تتيح لك أيضًا قيادة جهاز الكمبيوتر الخاص بك باستخدام الماوس. تشمل أمثلة أنظمة التشغيل التي تستخدم واجهة المستخدم الرسومية Windows و IBM's OS/2.

إذا كنت قد استخدمت نظام تشغيل قديمًا مثل الإصدار الأصلي من DOS، فلم تكن هناك واجهة مستخدم رسومية.

The advantages of using a GUI (Graphical User Interface)

All programs look similar and when you switch from a program supplied by one manufacturer to a different program supplied by another manufacturer, you will find the transition very easy.

Application programs work in the same way as the underlying operating system, and also look very similar, which means that they are easier to learn and use.

The GUI also allows programmers to easily write consistent-looking programs.

مزايا استخدام واجهة المستخدم الرسومية

تبدو جميع البرامج متشابهة، وعندما تنتقل من برنامج مقدم من شركة مصنعة إلى برنامج آخر مقدم من شركة مصنعة أخرى، ستجد أن الانتقال سهل للغاية.

تعمل برامج التطبيقات بنفس طريقة عمل نظام التشغيل الأساسي، كما أنها تبدو متشابهة جدًا، مما يعني أنها أسهل في التعلم والاستخدام.

كما تتيح واجهة المستخدم الرسومية للمبرمجين كتابة برامج ذات مظهر متسق بسهولة.

Part 3:
the internet and the web

الملزمة : 3

1.1 Concepts and terms

What is the difference between the internet and the WWW ?

-The internet relates to all the hardware and software involved in a computer's network . the **WWW** is a small part of this connection.

Internet Protocols :

HTTP: (Hypertext transfer protocol): It is the language that your web browser uses to request files or media . and it is shown at the beginning of a URL .

Hyperlinks: It is a piece of text that will do a function when clicked on. Like taking you to another website or section.

ISP (internet service provider) : It is the organization or a person that gives you access to the internet .

FTP(file transfer protocol): It is a way to transfer data from one place to another over the internet .

Websites and URLs:

A website is simply data stored on a WWW server and which can be accessed by people .

An URL is The special address of the website on the data base , which allow you to access the website .

Web browser: A tool that enables you to view web pages . like MS explorer .

Search Engine : it is a tool that you give a request or a prompt, and then searches the web for a websites that fit this prompt .

Cookies (website cookies): There are a small txt files that stores information about you for a specific website.

Internet Cache : A copy of the information of a website stored on the hard disk . it helps loading websites faster.

7.1.2.3 Encryption and it's uses

Encryption is the process of manipulating data to make it secure and privet .

The level of Encryption is often described by the **amount of bits used within the Encryption** . Like 128 bit Encryption is better than 32 bit Encryption . The more bits, the better.

7.1.2.4 & 7.1.2.5 The danger of surfing the web

There are multiple risks at browsing the internet . mainly , they are :

Viruses: Files that can harm your computer .

to counter it: use a An anti-Virus checker before downloading anything from the web.

Spam : Entering your email to unknown website can cause your email to be known for a bunch of websites and companies . which can be used to send you Ad-based emails.

To counter it: check the website if it is trusted or not .

Fraud : Entering your credit card for a unknown website can cause your credit card's information to be stolen . which can Lead to unwanted purchases from other people .

To counter it: make sure that the org you are dealing with has a good reputation .

7.1.2.6 Understand the term firewall.

A firewall consists of hardware and software protection against invasion over the internet .

Part 4:

Computer viruses and networks

الملزمة : 4

1. What is a Virus?

A **virus** is malicious software that infects systems by attaching to files or programs. It spreads through downloads, emails, or infected websites and can damage files, steal data, or disrupt operations.

الفيروس : هو نوع من البرامج الضارة (Malware) يمكنه نسخ نفسه والانتشار إلى برامج أو ملفات أخرى. غالبًا ما يسبب ضررًا للأجهزة من خلال حذف البيانات، أو إبطاء الأداء، أو سرقة المعلومات.

2.The Replication Routine

The replication routine is the process by which a virus makes copies of itself to spread across files, programs, or devices, often without the user's knowledge.

روتين التكرار هو العملية التي يقوم فيها الفيروس بإنشاء نسخ من نفسه لينتشر عبر الملفات أو البرامج أو الأجهزة، غالبًا دون علم المستخدم.

3. The Payload Routine

The payload routine is the part of a virus that performs the intended malicious action, such as deleting files, stealing data, or damaging system functions.

روتين الحمولة هو الجزء من الفيروس الذي ينفذ الهجوم الضار، مثل حذف الملفات، أو سرقة البيانات، أو إتلاف وظائف النظام.

4. The Network

A network is a group of connected devices (computers, servers, smartphones) that communicate and share resources through wired or wireless connection

الشبكة هي مجموعة من الأجهزة المتصلة (مثل الحواسيب والخوادم والهواتف الذكية) التي تتواصل مع بعضها البعض وتشارك الموارد عبر اتصالات سلكية أو لاسلكية.

5.Protocol

A **protocol** is a set of rules that define how data is transmitted and received over a network, ensuring devices can communicate properly. Examples include HTTP, FTP, and TCP/IP.

البروتوكول هو مجموعة من القواعد التي تحدد كيفية إرسال واستقبال البيانات عبر الشبكة، مما يضمن تواصل الأجهزة بشكل صحيح. من الأمثلة على ذلك: HTTP, FTP, TCP/IP.

6.TCP/IP (Transmission Control Protocol/Internet Protocol)

TCP/IP is a fundamental communication protocol used for sending and receiving data over the internet.

TCP ensures reliable data transfer, while **IP** handles addressing and routing the data between devices

TCP/IP هو البروتوكول الأساسي المستخدم في إرسال واستقبال البيانات عبر الإنترنت.

TCP يضمن نقل البيانات بشكل موثوق.

IP يتعامل مع عنوان البيانات وتوجيهها بين الأجهزة.